# 2025-26 Part III/MASt Astrophysics Research Project: Interim Report Coversheet

| **For completion by the student:** | | | | | |
|---|---|---|---|---|---|
| **Project Title:** | **Searching for Systematics in the Cosmic Microwave Background with Simulation-Based Inference** | | | | |
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| **Supervisor 1 Name:** | William Coulton | **Signature:** | William Coulton | **Date:** | 3/12/25 |
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# Searching for Systematics in the Cosmic Microwave Background with Simulation-Based Inference

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## 1 Background

As modern cosmological datasets have grown in complexity, it has become increasingly challenging to derive analytic probability distributions for key observables. Simulation-Based Inference (SBI) addresses this challenge by replacing explicit likelihoods with implicit distributions defined through simulations. However, the 'black box nature' of the machine-learning methods used in SBI requires the development of more validation methods supporting the reliability of resulting inferences.

This project aims to adapt and apply a suite of 'null and consistency tests' to assess the sensitivity of SBI analyses to systematic biases. We aim to develop these tests in the context of the Cosmic Microwave Background, implementing SBI for CMB power spectra. This involves constructing split datasets and introducing plausible systematics, then evaluating whether our SBI-null tests are able to detect the injected biases. Successful detection would increase confidence in SBI-based analysis. The methodology and implementation details are discussed further below.

This report presents the SBI pipeline implemented for this project and its application to a toy Lotka–Volterra model, which we use to develop and test divergence-based diagnostics for systematics. We then outline how these methods will be applied to CMB data in the full project. A flow chart describing the overall target workflow is shown in Figure 1 below.

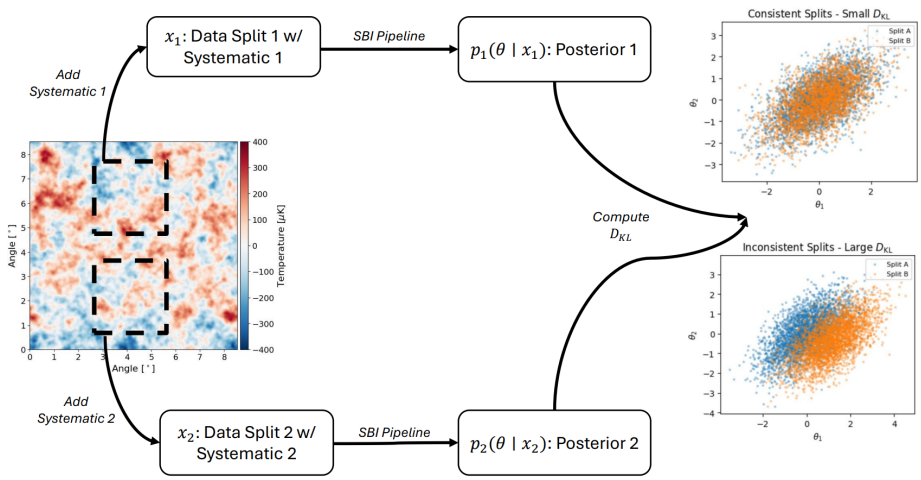


Figure 1: Overall workflow for the divergence-based diagnostic tests we aim to develop in this project.

## 2 The SBI Pipeline

We now outline the SBI framework implemented. The core SBI method chosen was neural posterior estimation (NPE). Once trained, NPE allows fast posterior sampling for many different data splits, which is essential for running repeated diagnostic tests. Importantly, NPE produces a normalised posterior estimator that can be both sampled from and evaluated pointwise, which is required for computing quantities such as KL divergences between posteriors. It is also worth noting that minimizing NPE loss is equivalent to maximizing mutual information between parameters $\theta$ and the learned summary network $T_\phi$, i.e. posterior estimates better match the true posterior for more sufficient data summaries. This provides another advantage to using NPE when working with CMB data, as the angular power spectra, $C_\ell$, provide near sufficient statistics for $\Lambda$CDM.

The first stage of the SBI pipeline is to define the simulator for the physical model. Observed data is then either generated from a known set of true parameters, or imported from external sources. This data is reduced to summary statistics, lowering dimensionality and reducing computational cost.

We then define a prior distribution over the parameters and draw $N$ samples $\theta_i \sim p(\theta)$. The parameters $\theta_i$ are inputted through our simulator, to produce data $x_i = f(\theta_i)$, forming a training dataset, $(\theta_i, x_i) \sim p(\theta)p(x|\theta)$.

At this stage, we implement our first diagnostic checks. We first conduct a coverage test, comparing the distribution of summary statistics for the training dataset with those of the observed data. This is illustrated in Figure 2. This

provides a qualitative check that the simulator, with a specified prior, can reproduce the observation, helping avoid model misspecification. We also detect and record the fraction of physically valid simulations. In cases of insufficient coverage or many invalid simulations, we adjust the model to ensure adequate support for the posterior estimator.

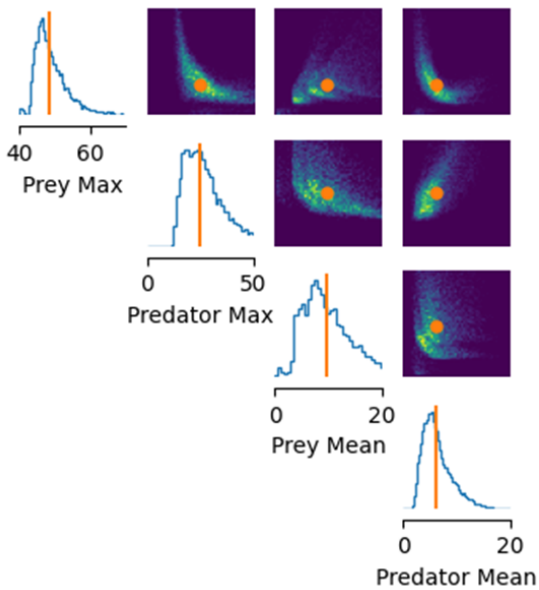


Figure 2: Coverage test comparing the distribution of simulated summary statistics (blue) with the observed summary statistics (red).

For a satisfactory prior, we then use our training dataset to train a neural density estimator, obtaining our posterior distribution. Utilising that our posterior estimator is amortized, we can efficiently plot the distribution of a large number of samples $\theta_i \sim p(\theta \mid x_{\text{obs}})$ from the posterior, comparing to the true values if known. This is illustrated in Figure 3. We then proceed to perform further diagnostic checks.

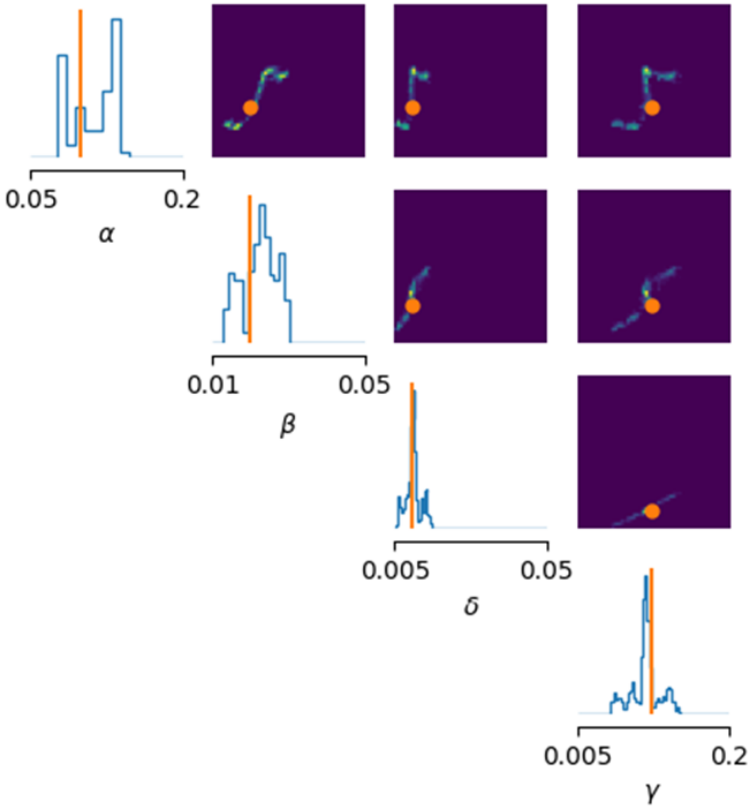


Figure 3: Posterior samples obtained from the neural posterior estimator for parameters $\alpha$, $\beta$, $\delta$, and $\gamma$. The orange lines indicate the ground-truth parameter values (for synthetic data).

The first diagnostic is a posterior predictive check. We draw samples $\theta_j \sim p(\theta \mid x_{\text{obs}})$ and use these samples with our simulator to generate predicted data $x_j^{\text{predictive}} = f(\theta_j)$. Plotting these predictive trajectories against the observed dataset provides a qualitative assessment of whether the inferred model can reproduce the observed dynamics.

We also perform simulation-based calibration (SBC) to assess the calibration of the inference procedure. This involves drawing samples from the prior distribution, simulating corresponding datasets, and inferring posterior samples for each dataset. The rank of the true parameter among the posterior samples is then recorded. For a correctly calibrated pipeline, these ranks are expected to be uniformly distributed. We therefore plot the empirical CDF of the ranks together with a 95% confidence interval under uniformity, providing a necessary but not sufficient check for calibration. An example SBC plot is shown in

Figure 4.

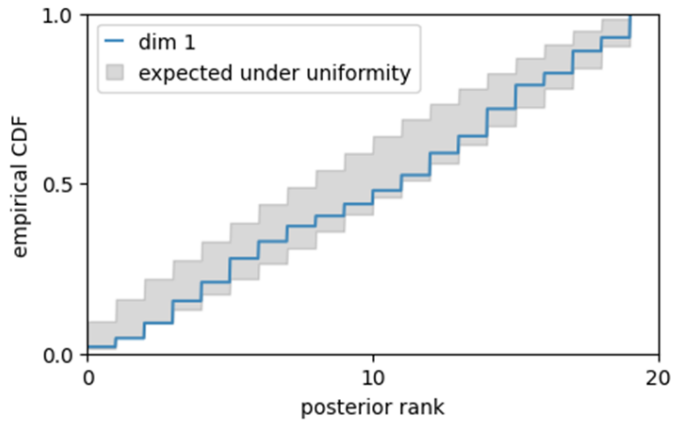


Figure 4: Simulation-Based Calibration (SBC) rank plot for one parameter dimension. The empirical CDF of ranks is compared against the expected uniform distribution. Deviations from the grey band indicate calibration issues or inference bias.

The implementation of this pipeline for a toy model is discussed in Section 3 below.

# 3 Implementing SBI – The Lotka-Volterra Model

The Lotka-Volterra predator-prey system, defined by the coupled differential equations

$$\frac{\mathrm{d}\,x}{\mathrm{d}t} = \alpha\,x \;-\; \beta\,x\,y, \tag{1}$$

$$\frac{\mathrm{d}\,y}{\mathrm{d}t} = \delta\,x\,y \;-\; \gamma\,y, \tag{2}$$

provided a simple model to test and develop our SBI pipeline. The figures included above were generated while working with this model, showing the output generated at each diagnostic stage. The summary statistics chosen for this model were the maximum and mean of the populations of predators and prey. A challenge initially encountered was that certain parameter ranges would produce unphysical behaviour, such as early prey extinction. This motivated the prior check for proportion of valid simulations, as previously discussed in Section 2. Having established a working SBI pipeline, we proceeded to implement data splitting and introduce tests for injected systematics. This process is discussed further in Section 4.

# 4 Implementing Data Splitting

Split datasets are obtained by running the simulator over a larger area, timespan, or volume. For example, in the Lotka-Volterra model, populations were simulated from the same set of initial conditions for double the usual simulation time, and then split into two halves. Noise can then be added to the summary statistics for each split, representing a plausible systematic. In the LV model, Poisson and Gaussian noise were added to each split, respectively. We are then able to run each dataset through our SBI pipeline, obtaining two separate posterior distributions, $p_1(\theta \mid x_{\text{obs}})$ and $p_2(\theta \mid x_{\text{obs}})$. Due to the presence of the injected systematic, we expect these posteriors to differ. To quantify this difference, we compute the Kullback-Leibler (KL) divergence,

$$D_{\text{KL}}(p_1 \,\|\, p_2) = \mathbb{E}_{p_1(\theta)} \left[ \log \frac{p_1(\theta)}{p_2(\theta)} \right]. \tag{3}$$

Note that we compute the KL divergence for each parameter individually in order to maintain a one-dimensional space. In higher dimensions, the regions of overlap between $p_1$ and $p_2$ decrease rapidly, increasing the likelihood that $p_2(\theta) = 0$ in regions where $p_1(\theta) > 0$, which results in $D_{\text{KL}} = \infty$. Computing the KL marginally avoids this instability.

These KL values provide a way of quantifying differences between the posteriors under the injected systematic. For example, in the Lotka-Volterra model, the KL divergence for each parameter was calculated for split datasets with different injected noise. The results are displayed in Table 1.

| **Parameter** | **KL Divergence** |
|---|---|
| $\alpha$ | 0.582 |
| $\beta$ | 0.339 |
| $\delta$ | 3.151 |
| $\gamma$ | 1.719 |

Table 1: KL divergence values for the marginal posteriors of each parameter using split datasets with differing injected noise.

To assess whether the divergence calculated is statistically significant, we plan to treat $D_{KL}$ as a test statistic and calibrate its distribution under a null hypothesis via Monte Carlo simulations. In Section 5, we discuss how similar methods can be adapted and applied within a cosmological context.

# 5 Next Steps: Adapting the Pipeline to CMB Data

The next stage of this project is to adapt the established SBI pipeline and accompanying diagnostic tests to Cosmic Microwave Background (CMB) datasets.

While the overall workflow will remain largely unchanged, several specific adjustments are required. A new cosmological simulator must be defined, with an appropriate set of input parameters to generate simulated CMB spectra. Suitable summary statistics must also be selected; the potential higher dimensionality of these summaries may also force adjustments to the training procedure. In addition, the strategy for data splitting must be reconsidered, along with the types of systematic effects and noise that can be meaningfully introduced into CMB observations.

Public libraries such as CAMB offer fast cosmological simulations, and the resulting angular power spectra $C_\ell$ provide near-sufficient summary statistics for $\Lambda$CDM analyses, as mentioned. Several approaches to data splitting can be explored, including comparing adjacent sky regions or observing the same region observed under different noise. There is also substantial flexibility in the types of systematics that can be introduced; examples include exploring different noise modelling choices or varying instrumental effects, such as beam profiles and gains. Finally, we plan to extend our suite of diagnostic tests, including evaluating the probability of one data split given another, as outlined in the original project description.